The Broadcast Ceiling: Information and Reliability Limits of Advantage Estimators in Long-Horizon RL

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Abstract

Long-horizon language-model RL often receives terminal trajectory rewards while updating token- or step-level actions. This paper studies when a scalar trajectory coefficient is enough and when reliable state-varying credit is needed. We decompose estimator error into an information term and a reliability term, proving a broadcast ceiling for any method whose coefficient is token-invariant under the available group context. CPU-only finite-MDP audits then compare group-relative broadcasts, prefix and structural baselines, sampled values, learned and oracle value TD, and policy-implied actor coefficients. The central empirical finding is conditional: trajectory-level coefficients can give high policy-gradient cosine, but they can be poor local credit explanations in heterogeneous traces; value-style TD reduces variance when coverage and observability are adequate; group methods can win when value estimates are blind or unsupported, while structural methods can recover part of the lost local information. The conclusion is not that PPO universally beats GRPO. Estimator choice should be made from credit heterogeneity, observability, support, drift, and compute cost.

Keywords. post-training; reinforcement learning from verifiable rewards; PPO; GRPO; advantage estimation; credit assignment; long-horizon agents.

1 Introduction

Long-horizon post-training for language-model agents is not merely long-context language modeling. A trajectory may include partial plans, tool calls, retrievals, edits, failed attempts, retries, compacted summaries, and final verifier feedback. In such traces, the question "did the rollout succeed?" is not the same as "which token or action helped?" A scalar terminal reward can be useful for ranking responses, but it can also broadcast the same update over helpful subgoals, harmful detours, and delay padding.

The method question is: *for variable-length, multi-step, tool-heavy trajectories, which information source can produce reliable token- or state-varying credit under a given compute budget?* We deliberately avoid centering the work on one base model or one training run. The target is a reusable mechanism study.

Contributions. We make five contributions. First, we formalize an information–reliability view of advantage estimation for long trajectories, including a generalized broadcast ceiling for token-invariant group context and a score-weighted counterpart tied to policy-gradient variance. Second, we separate policy optimization, group-relative normalization, policy-implied values, and on-policy distillation/self-distillation into a method taxonomy that treats OPD/OPSD as complementary post-training tools rather than direct replacements for PPO/GRPO. Third, we provide a CPU-only finite-MDP audit suite with exact behavior-policy advantages, analytic occupancy-measure policy gradients, stronger critic-free baselines, observability controls, and calibrated heterogeneity regimes. Fourth, we make estimator selection operational with a held-out audit-MSE plus cost rule on the phase grid. Fifth, we report mechanism findings across estimator fidelity, phase diagrams, length and token-cost sensitivity, tabular closed-loop training, and a tiny neural value-critic generalization audit.

Scope discipline. The paper makes a narrow claim about credit-assignment mechanisms. It does not claim that PPO is universally better than GRPO, that critic-free methods are trajectory-only, or that a finite-state toy predicts frontier-model behavior. We use the toy because the exact behavior-policy advantage is known, which lets us ask a question that is usually hidden in language-model training: whether an estimator is aligned with policy-relative action credit inside the trajectory. This is not the same as causal explanation. It is also not a claim that terminal-return REINFORCE is biased; REINFORCE remains an unbiased episodic policy-gradient estimator under the usual assumptions, but a trajectory-constant sample can localize action credit poorly and have high variance. A method can improve a benchmark for reasons unrelated to the mechanism under discussion, and a method can be mechanistically promising while still losing under a particular compute budget, implementation, reward model, or data mixture.

Organization. Section 2 defines the estimand and method families. Section 3 reviews related work. Section 4 gives compute and information costs. Section 5 proves the information–reliability decomposition and the broadcast ceiling. Sections 6–8 define the toy environment, experimental protocol, and results. Sections 9–11 interpret the mechanism evidence, compare method regimes, and state failure modes. The appendices provide implementation details and compact generated summaries; raw per-seed rows remain repository artifacts rather than submission-PDF content.

2 Definitions and Method Taxonomy

The target notation matters. Let the latent simulator state be $x_t \in \mathcal{X}$, the actor observation be $o_t = f_{\text{actor}}(x_t)$, and the critic observation be $z_t = f_{\text{critic}}(x_t)$. A sampled rollout follows a behavior policy

$$a_t \sim \pi_b(\cdot \mid o_t), \tag{1}$$

and produces a trajectory

$$\tau = (x_0, o_0, a_0, x_1, o_1, a_1, \ldots, x_T). \tag{2}$$

The outcome reward $R(\tau)$ may be combined with per-token KL, process rewards, or unit-test feedback. In the finite toy below, the exact comparison target is the behavior-policy action advantage:

$$Q^{\pi_b}(x, a) = \mathbb{E}_{\pi_b}[G_t \mid x_t = x, a_t = a], \tag{3}$$

$$V^{\pi_b}(x) = \mathbb{E}_{a\sim\pi_b(\cdot|f_{\text{actor}}(x))} Q^{\pi_b}(x, a), \tag{4}$$

$$A^{\pi_b}(x, a) = Q^{\pi_b}(x, a) - V^{\pi_b}(x). \tag{5}$$

The superscript is intentionally not $\star$: the paper does not measure alignment with an optimal-control advantage. It measures alignment with the exact on-policy/behavior-policy advantage under the rollout distribution used to generate the batch.

2.1 PPO with a critic

PPO optimizes a clipped policy-gradient surrogate [25]:

$$L^{\text{PPO}}(\theta) = \mathbb{E}_t \left[\min\left(\rho_t(\theta) A_t, \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon) A_t \right) \right], \tag{6}$$

where

$$\rho_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}. \tag{7}$$

In actor-critic use, a value model V_ϕ can estimate one-step temporal difference errors

$$\delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t), \tag{8}$$

or generalized advantage estimates [24]:

$$A_t^{\text{GAE}(\gamma,\lambda)} = \sum_{\ell=0}^{\infty} (\gamma\lambda)^\ell \delta_{t+\ell}. \tag{9}$$

The critic is valuable only if the value target is learnable, sufficiently covered, and cheaper than the credit it recovers.

2.2 GRPO

GRPO samples a group of completions for the same prompt and normalizes rewards inside the group [4, 26]. A simplified response-level advantage is

$$\widehat{A}_i^{\text{GRPO}} = \frac{R_i - \mu_G}{\sigma_G + \epsilon}, \tag{10}$$

where μ_G and σ_G are group statistics. This critic-free design is memory-efficient and effective in many verifiable-reward settings. Its long-horizon risk is intra-rollout information loss: one scalar can be broadcast across tokens with very different policy-relative action values.

2.3 Policy-implied values

VIMPO is a crucial nearby method because it is critic-free but not trajectory-constant [9]. In a KL-regularized derivation, its actor signal can be written in a token-varying form

$$\widehat{A}_t^{\text{VI}} = \beta \left[\log \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\text{ref}}(a_t \mid s_t)} - D_{\text{KL}}(\pi_\theta(\cdot \mid s_t) \,\|\, \pi_{\text{ref}}(\cdot \mid s_t)) \right]. \tag{11}$$

This signal escapes any theorem that applies only to trajectory-constant estimators. It also raises a reliability question: the identity is exact at a KL-regularized optimality condition, while practical training substitutes the current policy and a reference policy. Therefore a useful testbed should ask when the policy-implied signal is aligned with the current behavior-policy advantage and how that alignment changes with the fixed reference's KL distance from the current policy. The audit below evaluates this actor coefficient only; it does not implement the full VIMPO value-consistency loss or compare against the exact gradient of a matched KL-regularized objective.

Table 1: Training-signal taxonomy for long-horizon post-training. The important axis is whether a signal can vary within a trajectory and whether the information source is reliable. Distillation objectives are adjacent dense supervision, not direct policy-gradient advantage estimators.

Signal source	Representative methods	Unit	Reliability question
Trajectory outcome	GRPO; RLOO; REINFORCE	resp./group	Is group reward contrast informative enough despite the broadcast ceiling?
Learned value	PPO/GAE; VAPO; ArCHer	tok./state	Is the critic calibrated under the current policy distribution?
Sampled value	VinePPO; sampled MC	state/step	Is the branching budget large enough for low-variance values?
Structural evidence	SALT; GiGPO; anchor contrast	anchor	Are repeated anchors comparable and sufficiently covered?
Prefix evidence	BRPO; process rewards	prefix/step	Are intermediate verifier calls aligned with final return?
Policy-implied value	VIMPO	tok./state	Is the reference-policy identity reliable at this training stage?
Dense distillation	OPD; OPSD; dOPSD	token/step	Does teacher or future-context supervision match the reward objective?

2.4 Variance reduction versus credit assignment

The critic in actor–critic PPO does two jobs at once. It acts as a state-dependent baseline that can reduce policy-gradient variance, and it creates token- or state-indexed temporal credit. These are different mechanisms. A running global baseline can reduce variance without telling us which step helped; a structural group method can remain critic-free while recovering some step-level contrast; a sampled value method can estimate per-state credit without training a value network.

For a single token score $\psi_t = \nabla_\theta \log \pi_\theta(a_t \mid s_t)$, the baseline estimator has the form

$$\widehat{g}_t = (G_t - b_t)\psi_t. \tag{12}$$

Its variance is controlled by

$$\mathbb{E}\Big[(G_t - b_t)^2 \|\psi_t\|^2\Big], \tag{13}$$

not only by $\mathrm{Corr}(b_t, G_t)$ or $\mathrm{Var}(G_t - b_t)$. The variance-minimizing scalar baseline conditioned on state is

$$b^\star(s_t) = \frac{\mathbb{E}[G_t\|\psi_t\|^2 \mid s_t]}{\mathbb{E}[\|\psi_t\|^2 \mid s_t]}. \tag{14}$$

BRPO is useful in this framing because its prefix-budget rewards give a critic-free, position-aware baseline source rather than a general theorem that group baselines are only variance-reduction devices [22].

2.5 Exploratory reliability-gated baseline

The counterexample in this paper suggests a simple adaptive estimator. A critic should not be used merely because a value model exists; it should be used where the value estimate is locally supported. Let $n_D(s_t)$ be a replay or batch coverage count for the critic state abstraction, k a minimum support threshold, $\hat{A}_t^{\mathrm{TD}}$ a critic temporal-difference advantage, and $\hat{A}_i^G$ the group-relative scalar for trajectory i. Define

$$m_t = \mathbf{1}\{n_D(s_t) \geq k\}, \qquad \hat{A}_t^{\mathrm{CG}} = m_t \hat{A}_t^{\mathrm{TD}} + (1 - m_t)\hat{A}_i^G. \tag{15}$$

This reliability-gated estimator is intentionally modest. It does not solve critic blindness: if the state abstraction hides the causal feature, high coverage can still be misleading. Its purpose is to make critic readiness an explicit experimental variable and to preserve group-relative signal where the critic is locally undercovered.

Table 2: Two separable axes: how a baseline is formed and where credit is assigned. “SPO” is overloaded; here it refers separately to Single-stream Policy Optimization and Segment Policy Optimization.

Baseline source	Trajectory-level credit	Step-/segment-level credit
None	REINFORCE	–
Global	REINFORCE++; Single-stream PO	add process rewards
Sibling/LOO	GRPO; RLOO; DAPO; Dr. GRPO	GiGPO; SALT; anchor contrast; PRM+GRPO
Learned critic $V(s)$	usually unnecessary alone	PPO/GAE; ArCHer
Sampled value	–	VinePPO; OPPO; Segment PO
Policy-implied	–	VIMPO
Redistribution	–	RUDDER-style decomposition

3 Related Work

PPO and actor–critic post-training. PPO was introduced as a simpler clipped-objective alternative to trust-region policy optimization [25]. In RLHF-style language-model training, actor–critic optimization is usually wrapped with KL penalties, reference policies, reward models, old-logprob storage, reward normalization, and sequence-level preference signals [21, 27]. The relevant point for this work is not one historical RLHF recipe, but the availability of a state- or token-indexed value estimate. A critic can turn delayed outcome feedback into temporally structured advantages. That can help when rollouts contain subgoals and detours, but it also creates a second model that can drift, overfit, or assign confident false credit.

GRPO and RL with verifiable rewards. DeepSeekMath introduced GRPO as a critic-free method for mathematical reasoning, replacing the learned value model with group-relative reward normalization [26]. DeepSeek-R1 then made GRPO-style RL a central public reference point for reasoning-model training [4]. Subsequent work has analyzed the effective loss, success amplification, response-length effects, dynamic sampling, clipping details, group-relative bias, and statistical structure of GRPO-like gradients [17, 20, 23, 30, 31, 37]. The practical attraction is real: GRPO removes critic memory and critic training, which can be decisive at large scale. The risk studied here is that a response-level group scalar can be information-wasteful inside long, heterogeneous rollouts.

Baselines, sampled values, and step credit. Recent work shows that the design space is broader than PPO versus GRPO. REINFORCE-style and global-normalization methods reduce critic cost without necessarily solving intra-rollout credit [2, 8, 29]. VAPO is a particularly close value-based reasoning reference: it argues that long-CoT training needs reliable value estimation while explicitly addressing value bias, heterogeneous lengths, and sparse rewards [33]. VinePPO estimates values by branching Monte Carlo rollouts rather than fitting a value network [10]. ArCHer uses a hierarchical multi-turn value structure for language-model agents [38]. OPPO is a very recent token-credit proposal based on Bayesian value recursion and should be read as a promising taxonomy anchor rather than settled evidence [14]. RUDDER and Segment Policy Optimization connect the same question to delayed-reward redistribution and segment-level credit [3, 7].

Dense and structural step signals. Process supervision is another way to avoid a pure terminal broadcast. Stepwise verification and process reward models provide direct intermediate feedback [16, 28], while implicit process rewards try to recover denser signals from cheaper outcome supervision [32]. GiGPO and SALT are especially relevant because they preserve critic-free or group-based structure while constructing finer-grained comparisons across repeated states or trajectory graphs [5, 12]. These methods make the main point sharper: "critic-free" does not have to mean "trajectory-only," and "critic-based" does not automatically mean correct credit. Our anchor-action contrast row is a minimal local analogue of that structural idea for the toy environment; it is not a reproduction of GiGPO or SALT. Evidence-Calibrated Policy Optimization is especially close to our anchor-coverage and gating discussion: it shows that repeated-anchor evidence can be unreliable under limited rollout support and uses count shrinkage plus variance-gated critic-free credit on ALFWorld and WebShop [15]. Our novelty is therefore narrower: a broadcast-ceiling diagnostic, a finite-MDP phase diagram, and a cautious adaptive-selection framing.

Policy-implied and prefix-budget signals. VIMPO is the closest recent method to the taxonomy proposed here because it is critic-free while producing a token-varying signal from policy/reference log-ratios and a KL-regularized value identity [9]. Its existence is one reason this paper avoids the binary phrase "critic versus critic-free" as the main axis. The finite-MDP audit below treats VIMPO as a policy-implied estimator whose reliability depends on reference consistency and distance from the optimality condition. AnytimeReasoner and BRPO are also important because they use verifier calls at multiple thinking budgets to construct prefix-conditioned baselines [22]. We use their insight to report score-weighted residual-return diagnostics, while not claiming that a small MDP reproduces their full anytime-reasoning setup.

On-policy distillation. OPD and related generalized knowledge distillation methods train on student-visited states, reducing train/inference mismatch relative to purely offline teacher forcing [1, 13]. OPSD uses the same model in privileged teacher and ordinary student roles, aiming to capture dense reasoning supervision without a separate teacher model [36]. Recent OPD analyses emphasize teacher/student compatibility, teacher novelty, length inflation, and truncation collapse [13, 18]. These methods are close to the present problem because they also produce dense token-level signals on rollouts. They differ because their signal is a distillation target rather than a reward-improvement objective. Recent work extends on-policy self-distillation to diffusion LLMs. d-OPSD constructs a suffix-conditioned self-teacher from the model's own completed generation and applies divergence supervision at the denoising-step level, illustrating that the natural unit of dense supervision depends on the generation architecture [19]. Because this is a distillation objective rather than an advantage estimator, we treat it as adjacent rather than directly comparable work.

Long-horizon agentic training. Long-horizon agentic RL introduces issues that are often absent in short verifier tasks: tool-call state, file-system state, hidden evaluation artifacts, environment side effects, and compaction. The GLM-5.2 report is useful because it explicitly names compacted sub-traces, variable numbers of trainable traces per prompt, and a move from group-wise optimization to critic-based PPO with token-level advantages [34]. We treat this as a case study of a design pressure, not independent causal evidence that one optimizer family is universally superior. The 2026 credit-assignment survey [35] is useful context for why this paper is scoped as a diagnostic rather than a universal method claim: long-horizon reasoning and agentic RL span token, segment, step, turn, counterfactual, value-based, process-reward, and graph-based mechanisms.

4 Compute and Information Costs

Fair comparisons should charge all hidden inputs, not only final benchmark score. For PPO, the bill includes value-model parameters, value targets, critic forward passes, optimizer state, old-policy log probabilities, KL control, and rollout storage. For GRPO, the bill includes group size, discarded completions, low-dispersion groups, and verifier calls. For VIMPO, it includes reference-policy scoring and full-action KL approximations. For prefix-verifier or BRPO-style methods, it includes intermediate verifier calls. For structural methods, it includes matching repeated anchors or trajectory graphs. For dense distillation, it includes teacher or privileged-context compute, dense logits, trajectory length inflation, and truncation handling.

For benchmark reports, reward change should be reported beside its resource vector rather than treated as a context-free scalar:

$$\Delta\text{reward} \quad \text{given} \quad (\text{generated tokens},\ \text{wall clock},\ \text{extra model memory}), \tag{16}$$

reported alongside reproducible artifacts and negative results.

Policy and rollout cost. For PPO, each trainable token typically needs old-policy log probabilities, advantages, reference-policy information for KL control, and value-model targets. If rollouts are compacted into sub-traces, the implementation must decide whether every sub-trace is trainable, how truncation is handled, and how losses are normalized across very different lengths. GRPO avoids the critic but spends compute on group sampling. If many group members have identical verifier outcomes, the effective signal can be weak even when token generation cost is high.

Synchronization and straggler cost. Group-relative methods also have systems costs. AReaL documents the underutilization that arises when synchronous large-scale RL waits for long rollouts, and studies asynchronous training as a throughput/staleness tradeoff [6]. Straggler-aware group sizing studies the same pressure in synchronous GRPO-like training: larger groups can improve statistical baselines but increase the wall-clock risk that one long rollout stalls the group [11]. These are compute-cost issues, not direct credit-assignment theorems.

Distillation cost. OPD and OPSD can look cheap if only student updates are counted. A fair bill also includes teacher logits or hidden states, privileged traces, context construction, rejection/truncation policies, and any additional sampling needed to keep the student on-policy. Dense token supervision can be worth this cost, but it should not be compared to RL methods as if the dense signal were free.

5 Formal Limitation: Information, Reliability, and Broadcast

The title of this paper should not be read as “terminal-return policy gradients are biased.” A trajectory-level return can be a valid Monte Carlo coefficient for the episodic policy gradient:

$$\nabla J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta}\left[R(\tau) \sum_t \nabla_\theta \log \pi_\theta(a_t \mid o_t)\right]. \tag{17}$$

REINFORCE remains an unbiased estimator of this gradient under the usual on-policy assumptions. The issue studied here is different: a trajectory-constant scalar can be a poor sample-level localizer of policy-relative action credit inside a long heterogeneous rollout. It can reinforce helpful, harmful, and delay actions together in a single sampled trace, even though the expectation over many score vectors may still be correct for the total gradient.

Information–reliability decomposition. Let

$$Y = A^{\pi_b}(x_t, a_t) \tag{18}$$

be the target advantage for a sampled token. Let Z_m denote all information available to estimator family m: a trajectory return, sibling rollouts, critic state, sampled continuations, repeated anchors, prefix verifiers, or policy/reference log-ratios. For any estimator $\widehat{A}_m = h_m(Z_m)$,

$$\boxed{\mathbb{E}[(Y - \widehat{A}_m)^2] = \underbrace{\mathbb{E}[\mathrm{Var}(Y \mid Z_m)]}_{\text{information ceiling}} + \underbrace{\mathbb{E}[(\mathbb{E}[Y \mid Z_m] - h_m(Z_m))^2]}_{\text{reliability or consistency error}}} \tag{19}$$

by adding and subtracting $\mathbb{E}[Y \mid Z_m]$. The best possible squared correlation using the information source Z_m is

$$\max_h \mathrm{Corr}(Y, h(Z_m))^2 = \frac{\mathrm{Var}(\mathbb{E}[Y \mid Z_m])}{\mathrm{Var}(Y)}. \tag{20}$$

This is the core decision rule. A method helps when it reduces information error more than it increases reliability error and compute:

$$\widehat{m} = \arg\min_m \left\{ \widehat{\mathcal{I}}_m + \widehat{\mathcal{E}}_m + \lambda\, \mathrm{Cost}_m \right\}. \tag{21}$$

Trajectory-constant methods have a large $\widehat{\mathcal{I}}$ when within-rollout credit is heterogeneous. Critics have smaller information error when state is sufficient, but can have large fitting or calibration error. VIMPO has token-varying information, but its reliability depends on reference consistency and the KL-regularized optimality identity. Structural methods depend on repeated comparable states. Prefix methods depend on intermediate verifier quality.

Trajectory-broadcast ceiling. Sample a token pair (i, t) according to the weighting used for training. Unless otherwise stated, the experiments use token-uniform micro-averaging: each emitted token in the evaluation batch has equal weight. A trajectory-uniform or prompt-uniform convention changes the expectation but not the projection argument. Define

$$Y = A^{\pi_b}(x_{i,t}, a_{i,t}). \tag{22}$$

Let C_i denote all token-invariant information available to a broadcast method for rollout i: prompt identity, the sampled rollout, sibling rewards, group statistics, running baselines, and any other context that does not vary with token index t. This includes response-level GRPO, RLOO, global baselines, and prompt-group statistics. Consider all estimators of the form

$$\widehat{A}_{i,t} = g(C_i), \qquad t \notin C_i. \tag{23}$$

The L_2-optimal member of this class is

$$g^\star(C_i) = \mathbb{E}[Y \mid C_i], \tag{24}$$

and its irreducible error is

$$\min_g \mathbb{E}[(Y - g(C_i))^2] = \mathbb{E}[\mathrm{Var}(Y \mid C_i)]. \tag{25}$$

By the law of total variance,

$$\max_g \mathrm{Corr}(Y, g(C_i))^2 = \frac{\mathrm{Var}(\mathbb{E}[Y \mid C_i])}{\mathrm{Var}(Y)} = 1 - \frac{\mathbb{E}[\mathrm{Var}(Y \mid C_i)]}{\mathrm{Var}(Y)}. \tag{26}$$

Define the credit heterogeneity index

$$H_{\text{credit}} = \frac{\mathbb{E}[\operatorname{Var}(A^{\pi_b}_{i,t} \mid C_i)]}{\operatorname{Var}(A^{\pi_b}_t)}. \tag{27}$$

Then any token-invariant broadcast estimator has absolute token-level correlation at most

$$|\operatorname{Corr}(A^{\pi_b}_{i,t}, g(C_i))| \le \sqrt{1 - H_{\text{credit}}}. \tag{28}$$

This broadcast ceiling does not say that group or global methods cannot produce useful policy-gradient updates. It says that, when H_{credit} is large, any method that emits one scalar per trajectory has an information-theoretic token-fidelity ceiling. Learned critics, sampled values, process rewards, or structural repeated-state methods can beat that ceiling only if their own held-out reliability is high enough.

Score-weighted broadcast ceiling. Token correlation is not the whole optimization story because a token's contribution is also scaled by the policy score $\psi_{i,t} = \nabla_\theta \log \pi_\theta(a_{i,t} \mid o_{i,t})$. The score-weighted projection problem is

$$\min_g \mathbb{E}\left[\|\psi_{i,t}\|^2 \left(A^{\pi_b}_{i,t} - g(C_i)\right)^2\right]. \tag{29}$$

The optimal broadcast coefficient is

$$g^\star_\psi(C_i) = \frac{\mathbb{E}[\|\psi_{i,t}\|^2 A^{\pi_b}_{i,t} \mid C_i]}{\mathbb{E}[\|\psi_{i,t}\|^2 \mid C_i]}. \tag{30}$$

This is the formal reason that a trajectory-level method can have weak token-credit correlation yet good policy-gradient cosine: it may explain the score-weighted projection well even while failing as a local credit explanation. The experiments therefore report both token-level alignment and score-weighted residual diagnostics.

Operational diagnostic. The finite-MDP experiments report descriptive quantities side by side:

$$H_{\text{credit}} \quad \text{and} \quad R_{\text{critic}} = \operatorname{Corr}(\widehat{A}^V_t, A^{\pi_b}_t) \tag{31}$$

on held-out toy rollouts. Because R_{critic} directly uses the oracle target, it is not presented as an oracle-free deployment diagnostic. In real LLM training, H_{credit} should be estimated by a small branching audit, process labels, sampled values, or counterfactual rollouts. Critic reliability should be estimated by held-out value MSE, calibration slope/intercept, Bellman residuals, bootstrap or ensemble disagreement, policy-distribution drift, state-support counts, and rank correlation across independent continuations. The practical selection question is whether these observable proxies predict lower held-out estimator-selection regret.

6 Toy Experiment

6.1 Environment

The toy environment creates variable-length trajectories with three action types: `help`, `harm`, and `wait`. `help` increases a hidden progress score, `harm` decreases it, and `wait` leaves progress unchanged while consuming one decision step. It is therefore a *delay* action, not a true null action: even with zero explicit token cost, it can reduce future opportunity. A terminal verifier returns success when the final score reaches a prompt-specific threshold. Because the finite dynamics and behavior policy are known, we can compute the exact behavior-policy advantage:

$$A^{\pi_b}(x_t, a_t) = -c + V^{\pi_b}(x_{t+1}, h-1) - V^{\pi_b}(x_t, h), \tag{32}$$

where c is token cost and h is remaining horizon. Result artifacts keep the legacy field `oracle_advantage` for backward compatibility, but the estimand is A^{π_b}, not an optimal value advantage.

The exact-gradient audit uses a closely related finite MDP with four actions: `help`, `harm`, `delay`, and `null`. The `delay` action consumes effective decision horizon. The `null` action is a true irrelevant emission: it changes neither progress nor effective future opportunity. Its Bellman contribution is solved as a zero-reward self-loop, so its exact one-step advantage is zero. This lets the audit measure whether an estimator leaks credit onto irrelevant tokens separately from whether it penalizes opportunity-consuming delays.

Table 3: Observation access in the finite-MDP audits. The non-privileged critic uses $z_t = o_t$, matching the toy actor observation. Coarse and blind critics remove progress information to test reliability limits rather than privileged state access.

Variable	Actor	Non-priv. critic	Coarse critic	Blind critic
Prompt threshold	yes	yes	yes	yes
Progress score	exact	exact	sign only	no
Remaining horizon	yes	yes	yes	yes
Previous actions	no	no	no	no

6.2 Estimators and metrics

The group-relative estimator normalizes terminal returns inside each prompt group and broadcasts the resulting scalar to all tokens in the trajectory. The critic-style estimator fits a tabular value model from sampled returns-to-go and computes one-step temporal-difference advantages. We measure Pearson correlation with A^{π_b}, calibrated MSE after affine rescaling, raw MSE, sign accuracy, delay-action magnitude, within-trajectory variance, zero-variance group fraction, and exact critic-state hit rate.

The variance-credit grid adds eight non-exact estimators on the same toy dynamics: raw REINFORCE returns, a global return baseline, sibling group normalization, leave-one-out group centering, a dense progress-reward proxy, a critic-free anchor-action contrast over repeated state-action visits, a learned critic TD estimator, and a sampled Monte Carlo value estimator. The purpose is not to simulate every cited algorithm, but to test the mechanism classes in Table 2: baseline-only variance reduction, structural critic-free step credit, and value-estimation-based step credit.

Three robustness audits test likely objections. The anchor-coverage audit sweeps evaluation batch size to change how often exact state-action anchors repeat, testing whether the structural critic-free baseline is coverage-bound. The length-imbalance audit adds a length-adjusted group scalar, computed by normalizing return per token inside each prompt group, and sweeps maximum horizon. The token-cost audit reruns baseline and long-wait settings with per-token costs $c \in \{0, 0.01, 0.02, 0.05\}$, including the no-explicit-length-penalty case $c = 0$.

The phase-diagram audit uses the same environment but sweeps credit heterogeneity, critic observability, critic coverage, and reward contrast under a matched train/evaluation setting. The implementation can also run train/eval policy-shift cells, but the canonical paper grid keeps that axis fixed so the coverage and observability effects are isolated. For each cell it reports H_{credit}, the implied trajectory-broadcast ceiling, group correlation, critic reliability, and a crossover reading. This replaces a winner-count-only narrative with a method-selection diagnostic.

The closed-loop training audit uses the same toy dynamics but trains a tabular softmax policy with policy-gradient updates under matched generated-token budgets. This is still not full PPO or production GRPO: there is no neural policy, KL controller, minibatch optimizer, or clipped surrogate. It does test whether estimator-fidelity differences survive one downstream optimization loop. The compared signals are group total return, group return per token, critic TD from a replay value table, and the exploratory reliability-gated baseline.

The neural generalization audit keeps the policy fixed and replaces the tabular critic with a one-hidden-layer value network trained by SGD on thresholds 1 and 3, then evaluated on held-out threshold 2. Exact tabular state lookup cannot help in this setting because threshold is part of the exact state key. The audit therefore asks whether a small value model can interpolate enough structure to recover token-level temporal credit.

The behavior-policy-alignment matrix remains an estimator-fidelity experiment under controlled dynamics. The closed-loop audit is a downstream tabular-policy stress test, not a production PPO-vs-GRPO benchmark.

Exact policy-gradient audit. Because the MDP is finite, the exact-gradient variant computes the expected return $J(\theta)$ and the exact policy gradient $g^\star = \nabla J$ from occupancy measures and dynamic programming. Finite differences are reported only as an implementation check. For each estimator, we sample batches, form $\widehat{g} = \sum_t \widehat{A}_t \nabla_\theta \log \pi(a_t \mid s_t)$, and report relative mean-gradient error over independent replications, trace variance, cosine with $g^\star$, and one-step improvement under a matched policy-KL step. The estimator table includes sibling leave-one-out baselines, prefix-budget baselines, a BRPO-style combined baseline, cross-fitted learned value TD, and oracle-value TD. A separate policy-implied table reports VIMPO-style actor coefficients under equal and fixed-distance references.

7 Experimental Protocol

Design principle. The experiment is designed around a known behavior-policy target. In real language-model RL, we usually observe a reward and a model update but not the policy-relative action advantage. Here the finite-state dynamics allow exact dynamic programming for V^{π_b}, which turns the question into an estimator-fidelity problem. That lets us separate two claims that are often conflated: whether a method is cheap and whether its advantage signal points at the right actions under the rollout policy.

Case axes. The deep matrix varies five axes. The horizon axis tests whether scalar terminal broadcasts become less aligned as trajectories lengthen. The group-size axis tests whether more completions per prompt rescue GRPO-like normalization. The critic-budget axis tests coverage: a value table trained from too few groups should not be trusted. The observability axis removes or coarsens state features to simulate poor value-model inputs. The sparse-reward axis tests harder terminal feedback. Finally, the counterexample intentionally combines a blind critic and low coverage to create a regime where group-relative terminal outcomes should win.

Decision rule. The headline statistic is Pearson correlation with A^{π_b}, treated as a secondary interpretability metric rather than a full policy-gradient fidelity measure. A case is a mean critic win when the mean critic correlation exceeds the mean group-relative correlation. Because one mean win can be tiny, we also report a 95% across-seed confidence interval for the delta. A "clear critic" case has positive delta after subtracting the interval; a "clear group" case has negative delta after adding the interval; otherwise the case is labeled a near tie. The phase diagnostic uses an additional practical band of $|\Delta r| < 0.03$ before calling a crossover clear. This is why the abstract reports both the mean-win count and the more cautious confidence-interval reading.

Why not a full closed-loop PPO/GRPO benchmark? The closed-loop audit below is intentionally tabular and unclipped. A full PPO/GRPO benchmark would mix estimator fidelity with optimizer settings, clipping schedules, policy parameterization, exploration, and KL control. This paper therefore asks the behavior-policy credit question first, then uses a small downstream policy-training audit as a sanity check rather than as final production evidence.

8 Results

The canonical matrix uses 20 seeds and 18 fixed cases across horizon length, group size, critic budget, observability, sparse-reward regimes, and a designed counterexample. Critic-style TD has the higher mean correlation in 17 of 18 cases. A confidence-interval reading is more cautious: 16 cases are clearly critic-favorable, 1 is a near tie, and 1 is clearly group-favorable. The mean critic-minus-group correlation is 0.426. Table 4 reports representative cases; the full CSV and JSON are committed in `results/`.

Table 4: Selected rows from the 20-seed deep matrix. Delta is critic r minus group r; CI is a 95% across-seed interval for delta.

Case	Group r	Critic r	Delta	CI	Reading
horizon_04_baseline	0.495	0.977	0.482	±0.011	critic clear
horizon_16_long_wait	0.293	0.865	0.571	±0.013	critic clear
critic_budget_002_full	0.352	0.367	0.015	±0.027	near tie
critic_budget_128_full	0.352	0.906	0.554	±0.017	critic clear
observability_blind	0.353	0.482	0.129	±0.017	critic clear
blind_undercovered_counterexample	0.510	0.455	-0.055	±0.015	group clear
sparse_hard_group08	0.310	0.910	0.600	±0.008	critic clear

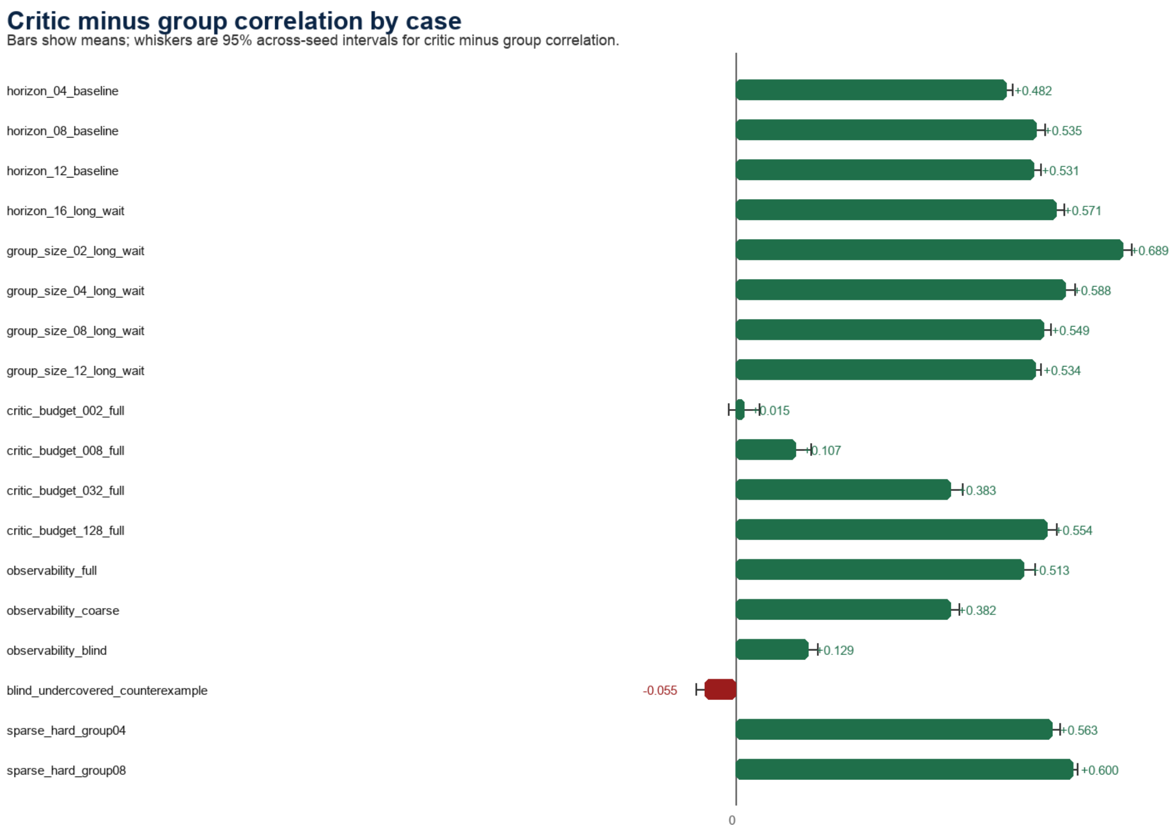

Figure 1: Mean critic-minus-group behavior-policy advantage correlation by case, with 95% across-seed confidence whiskers. Positive values favor critic-style temporal-difference advantages.

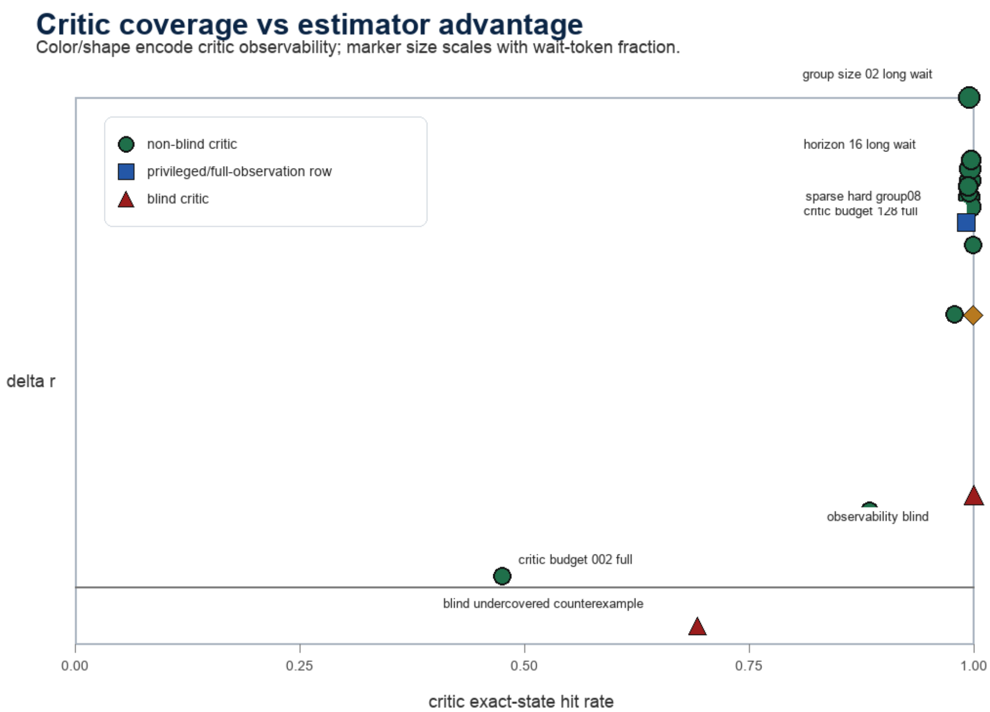

Figure 2: Critic state coverage versus estimator advantage. Marker color and shape encode critic observability; marker size scales with wait-token fraction. The group-relative counterexample occurs when the critic is blind and undercovered.

Interpretation. Within this controlled estimator-fidelity toy, the evidence supports the state-conditioned value-TD hypothesis when progress state is observable and critic coverage is adequate: value-style temporal-difference advantages track policy-relative action credit more closely than response-level group scalars. PPO is one implementation route for such a signal, not the object isolated by this experiment. The result also protects against overstatement. In the coverage-limited case, the confidence interval crosses zero; in the blind and undercovered case, group-relative terminal outcome information is better than a poor value table.

Table 5: Exact finite-MDP policy-gradient fidelity over replicated batches. The target gradient $g^\star$ is computed by occupancy-measure dynamic programming; finite differences are used only as a check. Err. is $\|\overline{\widehat{g}} - g^\star\|/\|g^\star\|$, Var. is across-batch trace variance, Cos. is gradient cosine with $g^\star$, ΔJ_b is per-batch matched-KL improvement, Neg. is the probability of a negative per-batch update, and Null $|\hat{A}|$ is estimated absolute credit on the true null action.

Estimator	Err.	Var.	Cos.	$\overline{\Delta J_b}$	P5 ΔJ_b	Neg.	Null $\lvert\hat{A}\rvert$
REINFORCE	0.075±0.020	0.0182±0.0009	0.997±0.010	0.156	0.132	0.000	0.610
Sibling LOO	0.050±0.015	0.0129±0.0006	0.999±0.007	0.163	0.146	0.000	0.471
Prefix baseline	0.049±0.013	0.0101±0.0004	0.999±0.006	0.168	0.151	0.000	0.493
BRPO-style	0.048±0.014	0.0108±0.0005	0.999±0.006	0.167	0.150	0.000	0.488
Learned value TD	0.043±0.014	0.0062±0.0004	0.999±0.007	0.175	0.158	0.000	0.000
Oracle-value TD	0.026±0.009	0.0021±0.0001	1.000±0.002	0.186	0.176	0.000	0.000

$\|g^\star\| = 0.160$; base $R = 0.506$; 200 replications. ΔJ_b: per-batch matched-KL.

Table 6: Policy-implied actor-coefficient alignment. These rows evaluate the VIMPO-style actor coefficient against the unregularized behavior-policy gradient target used in this toy. They are not full VIMPO updates and should not be read as algorithm-level performance.

VIMPO-style diagnostic	Ref. KL	Err./$\|g^\star\|$	Cos.	ΔJ	r_A	Reading
$\pi = \pi_{\text{ref}}$	0.000	1.000	0.000	0.000	0.000	actor coefficient is zero at equal reference
fixed ref., near	0.003	0.896	0.659	0.124	0.584	chosen path: direction fixed, magnitude mismatch changes
fixed ref., mid	0.018	0.786	0.659	0.124	0.584	chosen path: direction fixed, magnitude mismatch changes
fixed ref., far	0.060	0.764	0.659	0.124	0.584	chosen path: direction fixed; not full VIMPO performance

Rows are policy-implied actor-coefficient alignment diagnostics, not complete VIMPO algorithm performance or KL-regularized objective comparisons.

Table 7: Prefix and baseline diagnostics from the exact-gradient audit. The score-weighted residual moment is $\mathbb{E}[(G_t - b_t)^2 \|\nabla \log \pi(a_t \mid s_t)\|^2]$, the quantity most directly related to single-token score-function variance.

Baseline	Corr(b_t, G_t)	Var$(G-b)$/Var(G)	Score-wtd. resid.	Count
Group mean	-0.011	1.256	0.194	85920
Prefix budget	0.055	1.000	0.155	85920
BRPO-style	-0.011	1.067	0.165	85920
Critic value	0.460	0.789	0.126	85920

Exact-gradient audit. Tables 5–7 are the main upgrade over correlation-only evidence. REINFORCE, sibling leave-one-out, and prefix baselines have excellent gradient cosine with the exact policy gradient even though their token-level advantage correlations are modest. That is expected: terminal returns can be valid Monte Carlo policy-gradient coefficients while still being weak local credit explanations. Oracle-value TD is the dynamic-programming upper bound and has the lowest trace variance and lowest null-action leakage in this covered tabular setting; the cross-fitted learned-value row retains most of the token-credit correlation while exposing finite-sample coverage risk. The table reports intervals over replicated batches and per-batch matched-KL improvement statistics, so variance can show up as lower fifth-percentile improvement or higher negative-update probability rather than only as a mean-gradient direction. Table 6 is deliberately separate. It evaluates only the VIMPO-style actor coefficient. At equal reference the coefficient is zero for the actor term; along the chosen fixed-reference path, the nonzero-KL rows have nearly fixed direction while their magnitude mismatch changes. This is an objective-mismatched diagnostic rather than a full VIMPO result. The baseline table gives the analogous BRPO-style message: prefix and combined baselines can reduce score-weighted residual moments relative to a group mean, but the critic value baseline remains strongest in this supported toy.

Table 8: Broadcast-ceiling phase diagnostic over credit heterogeneity, observability, critic coverage, and reward contrast under matched train/evaluation dynamics. The ceiling is $\sqrt{1 - H_{\text{credit}}}$, the best possible absolute correlation for a token-invariant broadcast estimator under the reported token weighting.

Regime	H_{credit}	Ceiling	Group r	Critic r	Δr	Crossover reading
realized H=0.00 non-priv high cov.	0.000	1.000	0.901	1.000	0.099	critic clear; either; choose by cost
realized H=0.25 non-priv high cov.	0.246	0.868	0.665	0.999	0.334	critic clear; either; choose by cost
realized H=0.25 non-priv low cov.	0.251	0.865	0.730	0.788	0.058	near tie; either; choose by cost
realized H=0.50 non-priv high cov.	0.497	0.709	0.624	0.997	0.373	critic clear; critic/sampled value
realized H=0.70 blind low cov.	0.704	0.543	0.410	0.285	-0.125	group clear; process/structural/hybrid
realized H=0.84 blind low cov.	0.843	0.396	0.301	0.189	-0.112	group clear; process/structural/hybrid
48 cells; 11 critic clear, 3 group clear, 34 near ties.						

Broadcast-ceiling phase diagram. Table 8 is the main response to the boundary critique. In the 48-cell calibrated phase grid, realized H_{credit} spans from near-zero homogeneous traces to roughly 0.88 heterogeneous traces. Critic coverage is controlled by a fixed training-trajectory budget, so eval sibling group size does not secretly increase critic data. High-support non-privileged critics produce clear critic-favorable crossovers, while blind or low-support cells produce group-favorable and near-tie regimes. The stronger working claim is therefore conditional: critic benefit is predicted jointly by within-trajectory credit heterogeneity and held-out critic reliability.

Estimator-selection regret. Table 9 operationalizes the selection rule rather than leaving it as a proposal. The rule uses diagnostic-batch MSE estimates plus a fitted critic-cost penalty, then selects group or critic on held-out phase-grid cells. This audit is intentionally small and only compares the two estimators available in the phase grid, but it tests the practical claim that information and

Table 9: Held-out estimator-selection regret on the phase grid. The audit-MSE+cost policy fits a scalar critic-cost penalty on half the cells. In each cell, alternating seeds split audit MSE estimates from held-out regret measurement. Regret is held-out calibrated MSE above the oracle best of group and critic for that cell.

Policy	Mean regret	Max regret	Accuracy	Critic rate
Audit MSE+cost	0.00031	0.00745	0.958	0.292
Always group	0.00566	0.07478	0.750	0.000
Always critic	0.00234	0.01666	0.250	1.000
24 train cells, 24 held-out cells; fitted $\lambda = 0.0005$.				

reliability diagnostics can reduce selection regret relative to static always-group or always-critic policies.

Table 10: Variance-reduction versus credit-assignment grid on the long-wait toy case. Baseline-only methods can lower second moment, but they do not create within-trajectory credit variation. Abbreviations: Var. is variance reduction; Group z is sibling-group z-scoring; LOO is leave-one-out; Within is within-trajectory estimator variance; Anchor ctr. is anchor-action contrast.

Estimator	Var.	Credit	r	MSE	Wait	Within
REINFORCE	none	traj.	0.357	0.0208	1.157	0.000
Global	global	traj.	0.357	0.0208	0.875	0.000
Group z	group	traj.	0.319	0.0214	0.855	0.000
LOO	group	traj.	0.332	0.0212	0.799	0.000
Dense reward	dense	step	0.674	0.0130	0.000	0.416
Anchor ctr.	anchor	step	0.715	0.0117	0.439	0.042
Critic TD	critic	step	0.870	0.0058	0.356	0.031
Sampled MC	sampled	step	0.849	0.0067	0.563	0.034

Variance-credit decomposition. Table 10 makes the decomposition explicit. The global baseline reduces the second moment of the trajectory-level policy gradient signal relative to raw REINFORCE returns, but it leaves within-trajectory variance at zero because every token in a trajectory still receives the same scalar. Sibling group normalization is memory-light and critic-free, but in this long-wait setting it also remains a trajectory-level broadcast. The anchor-action contrast row tests a stronger critic-free structural baseline: it uses repeated state-action anchors to create local step-level comparisons without fitting a value model. It is a transductive batch estimator over exact repeated toy states: it assumes repeated state-action coverage, and unsupported anchors receive zero rather than a borrowed value estimate. It substantially narrows the gap to the critic rows, improving correlation from the sibling-group baseline's 0.319 to 0.715, but it remains below learned critic TD at 0.870 and sampled Monte Carlo value estimation at 0.849. The dense progress-reward proxy also gives a step signal, but because it lacks horizon and threshold information, it is not a full value estimate. The strongest claim supported by this grid is therefore not that every critic-free method is weak; it is that long-horizon tasks need some mechanism for state-sensitive step credit, and in this toy the value-estimation mechanisms still carry the highest behavior-target alignment.

Anchor coverage audit. Table 11 checks the assumption behind the structural critic-free baseline. At two evaluation groups, only 41.3% of steps have supported repeated anchors and anchor contrast is worse than sibling grouping ($r = 0.134$ versus $r = 0.361$). The method crosses the sibling baseline at eight groups, reaches support above 0.80 at 16 groups, and climbs to $r = 0.751$

Table 11: Anchor-coverage audit over five seeds. Groups is the number of evaluation prompt groups in the estimator batch; Support is the fraction of steps with leave-one-out state-action anchor support. $A - G$ is anchor r minus group r; $C - A$ is critic r minus anchor r.

Groups	Support	Group r	Anchor r	Critic r	$A - G$	$C - A$
2	0.413	0.361	0.134	0.805	-0.228	0.672
4	0.510	0.341	0.253	0.836	-0.088	0.583
8	0.675	0.329	0.389	0.852	0.060	0.463
16	0.813	0.326	0.544	0.845	0.219	0.301
32	0.915	0.321	0.654	0.861	0.334	0.206
48	0.945	0.318	0.710	0.867	0.392	0.156
64	0.962	0.315	0.751	0.865	0.437	0.113

at 64 groups. Critic TD remains higher in every row. This is a useful negative and positive result at once: structural critic-free step credit can be real, but it is not free. It needs repeated comparable states, and sparse-anchor regimes should fall back to group or critic signals rather than pretending a local contrast exists.

Table 12: Length-imbalance audit on the long-wait scenario. Len gap is the mean within-group max-minus-min trajectory length. Len-adj. is a group scalar based on return per token; Δ_G and Δ_L are critic r minus group-total and length-adjusted group r.

$H_{\max}$	Len gap	Wait	Group r	Len-adj. r	Critic r	Δ_G	Δ_L
4	1.775	0.336	0.488	0.500	0.981	0.492	0.481
8	4.851	0.431	0.389	0.389	0.940	0.551	0.551
12	7.725	0.495	0.330	0.328	0.890	0.561	0.562
16	10.696	0.539	0.293	0.289	0.849	0.556	0.560
20	13.439	0.573	0.266	0.260	0.811	0.545	0.551

Length-imbalance audit. Table 12 tests whether a simple length correction rescues trajectory-level grouping. It does not in this toy. As $H_{\max}$ increases from 4 to 20, mean within-group length range grows from 1.775 to 13.439 tokens and delay-action fraction grows from 0.336 to 0.573. The length-adjusted group scalar remains close to the total-return group scalar: at $H_{\max} = 20$, critic r is 0.811, group-total r is 0.266, and length-adjusted group r is 0.260. Length correction changes the rollout ranking signal, but it still broadcasts one number to all tokens.

Table 13: Token-cost sensitivity over 20 seeds. The $c = 0$ rows remove the explicit per-token penalty. CI is a 95% interval for critic-minus-group correlation; Wait_G and Wait_C are delay-action magnitude ratios.

Scenario	Cost	Group r	Critic r	Δ	CI	Wait_G	Wait_C
base	0.00	0.336	0.882	0.546	0.010	0.930	0.465
base	0.01	0.332	0.882	0.550	0.010	0.937	0.465
base	0.02	0.335	0.882	0.546	0.010	0.928	0.465
base	0.05	0.339	0.882	0.543	0.010	0.902	0.465
long	0.00	0.335	0.888	0.552	0.015	0.871	0.371
long	0.01	0.327	0.888	0.561	0.016	0.911	0.371
long	0.02	0.330	0.888	0.557	0.016	0.898	0.371
long	0.05	0.334	0.888	0.554	0.016	0.868	0.371

Token-cost sensitivity. Table 13 checks whether the critic result is just a side effect of the per-token cost. The answer is no for these controlled dynamics. All eight scenario/cost rows have positive critic-minus-group correlation after subtracting the 95% interval. In the long-wait, zero-cost row, critic r is 0.888, group r is 0.335, and the delta is 0.552. Removing the explicit length penalty therefore does not remove the temporal-credit gap.

Table 14: Closed-loop tabular policy training under matched generated-token budgets. Default uses the standard replay budget; low cov. restricts replay and raises the reliability gate. Cov-gated is an exploratory reliability-gated baseline.

Setting	Method	Init R	Final R	ΔR	Success	Critic frac.
default	Group	0.515	0.785	0.271	0.949	0.000
default	Len-group	0.515	0.782	0.267	0.945	0.000
default	Critic TD	0.515	0.810	0.296	0.974	0.998
default	Cov-gated	0.515	0.809	0.294	0.972	0.996
low cov.	Group	0.508	0.769	0.261	0.928	0.000
low cov.	Len-group	0.508	0.754	0.246	0.913	0.000
low cov.	Critic TD	0.508	0.780	0.272	0.940	0.945
low cov.	Cov-gated	0.508	0.777	0.269	0.937	0.644

Closed-loop training. Table 14 is the first downstream optimization check. The result is more conservative than the estimator-fidelity tables, as expected. In the default setting, group-total training improves mean return from 0.515 to 0.785, while critic TD reaches 0.810 and the reliability-gated baseline reaches 0.809. In the low-coverage stress setting, group total reaches 0.769, critic TD reaches 0.780, and the reliability-gated baseline reaches 0.777 while using critic credit on only 64.4% of final-batch steps. The adaptive estimator is therefore a candidate design pattern, not a claimed replacement for PPO or GRPO: it preserves most critic benefit while making critic use explicit and auditable.

Table 15: Tiny neural value-critic generalization audit. The value critic is trained on thresholds 1 and 3, then evaluated on held-out threshold 2. Exact tabular state lookup cannot match the evaluation threshold, so the TD signal comes from feature interpolation.

Estimator	r	MSE	Sign	Wait	Within
Group	0.364	0.0318	0.674	0.917	0.000
Neural TD	0.842	0.0104	0.875	0.418	0.038
Eval keys unseen in tabular train: 1.000; neural minus group r: 0.479.					

Tiny neural generalization. Table 15 is not an LLM-scale experiment, but it does remove the most brittle tabular assumption. On the held-out threshold, the group-relative broadcast reaches Pearson $r = 0.364$ against A^{π_b} token credit, while neural critic TD reaches $r = 0.842$ with lower calibrated MSE and lower delay-action magnitude. This strengthens the mechanism claim, while still leaving a larger transformer or nanoGPT-scale benchmark as future work.

9 Mechanism Analysis

Horizon length. The horizon cases show the central intuition. As trajectories lengthen and wait tokens accumulate, a response-level scalar is asked to explain increasingly heterogeneous token roles.

The group-relative estimator can still rank complete rollouts, but that does not mean it assigns good token credit. The critic-style estimator retains access to progress state and remaining horizon, so it can separate helpful actions from delay actions even when the final reward is delayed.

Group size. Increasing group size improves the quality of group statistics only when the additional samples create meaningful reward dispersion. In the long-wait cases, larger groups help the group-relative estimator somewhat, but the broadcast problem remains: every token in a completion inherits the same normalized outcome. In settings where verifier outcomes are sparse or clustered, larger groups can spend many tokens for little additional advantage information.

Critic budget and coverage. The critic-budget axis is the strongest warning against a simplistic "critics are better" claim. With only two training groups, the critic has an exact-state hit rate below one half and the confidence interval crosses zero. With more coverage, the critic delta grows sharply. In a neural setting, the analogous quantity is not table hit rate but value-model calibration under the current policy distribution. A poorly covered critic should be treated as a risk, not a reason for confidence.

Observability. The observability cases show that a critic can still extract some value from coarse state, but a blind critic loses much of the temporal structure. The toy blind critic is intentionally simple; real long-horizon agents can become effectively blind for subtler reasons, including lossy compaction, missing tool state, stale memory, hidden environment variables, or value-model inputs that do not include the causal feature needed for future return.

Counterexample. The counterexample is the scientific control. In the blind-undercovered case, terminal group outcomes carry more usable information than a weak value table. The resulting group win is small but clear under the confidence-interval rule.

10 Estimator Regimes and Selection Criteria

The answer is conditional, and the condition is about information and reliability. Critic methods are more plausible when future return is predictable from prefix state, rollouts contain mixed token roles, and held-out value calibration is good enough to justify the extra model. Group-relative methods are more plausible when rewards are cheap, verifiable, and meaningfully dispersed within prompt groups, or when the critic would be blind, stale, or too expensive. Prefix-verifier and structural methods are plausible middle ground when they can create step-varying evidence without fitting a full critic. VIMPO-like policy-implied methods are plausible when the reference policy and KL-regularized value identity are reliable enough for the current training stage.

The practical selection rule is therefore not "use a critic" or "avoid a critic." It is to estimate the information ceiling, the reliability error, and the resource vector. Report generated tokens, verifier calls, reference-policy scoring, critic memory, sampled branches, dense-logit cost, synchronization stalls, and wall-clock time beside reward changes.

11 Failure Modes and Testable Predictions

PPO/critic failures. Critics can overfit reward artifacts, become stale after policy drift, collapse on long sparse trajectories, or assign confident but wrong temporal credit.

GRPO failures. GRPO can suffer zero-variance groups, low-dispersion amplification, response length bias, scalar reward broadcast over harmful tokens, and group composition confounding prompt difficulty with policy quality. These issues appear in analyses of effective loss, R1-Zero-like training, DAPO-style sampling, and group-relative bias [17, 20, 31]. Related theoretical and stability analyses study group-relative bias and MDP-level corrections [23, 30, 37].

Dense distillation failures. Teacher and student reasoning patterns may be incompatible. The teacher may add little new information, on-policy rollouts may inflate in length, and privileged self-teacher traces may leak solution structure [13, 18]. Dense self-distillation variants can also fail when architecture-specific teacher signals mismatch the reward target [19, 36].

Predictions. We predict that increasing horizon length and delay padding will reduce the Pearson alignment between response-level group advantages and exact or process labels. We predict that increasing group size will help GRPO mainly when it increases within-prompt reward dispersion. We predict that adding process rewards will narrow the PPO/GRPO credit gap, and that degrading critic observability will produce regimes where GRPO wins. Sampled-value methods should approach critic-like credit when the branching budget is high enough, but their inference cost should dominate in long agentic traces unless branch reuse or state sharing is available.

12 Threats to Validity

Toy dynamics. The toy has a small finite state and a known exact target. Most audits use tabular estimators, and the neural audit uses only a tiny value network on held-out toy thresholds. This is deliberately unlike frontier post-training. The benefit is interpretability; the cost is that transformer-scale generalization, distribution shift, optimizer dynamics, and reward-model error are not measured.

Estimator versus optimizer. The experiment compares advantage estimators, not complete PPO and GRPO training loops. A full optimizer could amplify or dampen estimator differences through clipping, KL schedules, entropy, minibatching, and policy parameterization. This paper therefore supports a mechanism hypothesis, not a leaderboard conclusion.

Simplified GRPO baseline. The group-relative baseline is intentionally simple. Future comparisons should include leave-one-out baselines, length-normalized objectives, Dr. GRPO-style corrections, DAPO-style dynamic sampling, token-level process rewards, and low-dispersion safeguards. A stronger GRPO family member may narrow the gap in some regimes.

Simplified critic. The main critic is tabular and trained from returns-to-go. The tiny neural audit checks interpolation across one held-out threshold, but it is still far from a large value model trained on language traces. Neural value models can generalize across states, but they can also hallucinate value structure. The toy's exact-state hit rate is a proxy for coverage, not a complete model of critic reliability.

13 Reproducibility

The experiment code is CPU-only and uses no external Python dependencies. The repository contains canonical JSON, CSV, Markdown, and generated-table artifacts for every result reported here, plus unit tests, a smoke runner, and a LaTeX build script. The public PDF is generated from those artifacts rather than hand-copied tables. Raw seed-level rows are intentionally kept as repository artifacts so that the submission PDF reports aggregate evidence without becoming a run ledger.

14 Limitations

The toy environment has a finite known state and a tabular critic. Real post-training adds neural approximation, KL controllers, reward-model errors, tool failures, distributed rollout systems, stale policies, optimizer details, and data contamination risks. The GRPO baseline is intentionally simple, and the VIMPO audit reports actor-coefficient alignment rather than a full algorithm. A stronger external-validity claim requires a controlled sequence-policy benchmark with clipped objectives, KL control, corrected GRPO-family variants, process rewards, and matched rollout budgets.

15 Conclusion

For long heterogeneous trajectories, value-based temporal estimators are a strong candidate when value targets are learnable and adequately covered; critic-based PPO is one implementation of that broader design choice. GRPO remains attractive when group reward contrast is reliable and critic modeling is too costly or poorly observed. VIMPO-like policy-implied signals, BRPO-like prefix baselines, structural anchors, sampled values, and process rewards all occupy meaningful middle ground. The paper's central claim is therefore not "PPO always beats GRPO," but a set of falsifiable boundaries for when token- or state-varying credit is worth its reliability and compute cost.

A Formal Objective Notes

Broadcasted group advantages. In a response-level group-relative update, a completion-level reward statistic is often applied to every trainable token in that completion. Let $g_i = (R_i - \mu_G)/(\sigma_G+\epsilon)$ be the group-normalized scalar for completion i. The token update is then proportional to $g_i \nabla_\theta \log \pi_\theta(a_{i,t} \mid s_{i,t})$ for many t. This is efficient, but in a single sampled completion it reinforces all sampled actions with the same scalar. That sample-level localization is weak whenever success depends on a small subset of steps and the rest are helpful delay, harmful detours, or irrelevant formatting. This does not contradict the unbiasedness of the total REINFORCE gradient; it states a token-fidelity limitation of trajectory-constant coefficients.

Temporal-difference advantages. A critic-style estimator replaces one response-level scalar with a local difference between predicted future return before and after an action. In the toy, this is a tabular one-step TD estimate. In language-model agents, the same idea may be implemented with value heads, separate value models, process reward models, or learned subgoal predictors. The central requirement is that the state representation contains enough information to make future return predictable.

B Exact Gradient Derivation

The exact-gradient audit uses a finite MDP with a true null self-loop. Let $s = (q, h)$ denote progress score q and remaining effective horizon h. For $h > 0$, the non-null actions transition to smaller horizon, while the null action returns to the same state with zero reward. The value equation is

$$V^\pi(s) = \sum_{a \neq \mathrm{null}} \frac{\pi(a \mid s)}{1 - \pi(\mathrm{null} \mid s)} \left[r(s,a) + V^\pi(s') \right], \tag{33}$$

which is the closed-form solution of the null self-loop. This keeps the null action's one-step advantage exactly zero:

$$A^\pi(s, \mathrm{null}) = 0 + V^\pi(s) - V^\pi(s) = 0. \tag{34}$$

Let $\eta^\pi(s)$ be the expected number of visits to state s before effective horizon reaches zero. Because null actions can repeat without advancing horizon, the implementation propagates incoming non-null mass and divides by $1 - \pi(\mathrm{null} \mid s)$ to obtain total expected visits. The policy gradient for the unregularized expected return is then

$$\nabla_\theta J(\theta) = \sum_s \eta^{\pi_\theta}(s) \sum_a \pi_\theta(a \mid s) A^{\pi_\theta}(s,a) \nabla_\theta \log \pi_\theta(a \mid s). \tag{35}$$

This is the quantity reported as $g^\star$ in Table 5. The central-difference gradient is not used as the target; it is only a check that the dynamic-programming implementation and the softmax parameter indexing agree. In the canonical run, the relative finite-difference check error is approximately 2.0×10^{-10}.

The reported "error" column in Table 5 is therefore not a theorem-level estimator bias unless the expectation is enumerated exactly. It is the norm of the empirical mean gradient error across the replicated batch experiment:

$$\frac{\|\overline{\widehat{g}} - g^\star\|}{\|g^\star\|}. \tag{36}$$

REINFORCE-style coefficients can be unbiased in expectation and still show a nonzero finite-replication mean error. This is why the table reports replication intervals and trace variance beside token-credit correlation.

C Objective-Mismatch Note for Policy-Implied Values

The VIMPO-style rows in this paper are intentionally separated from the main estimator table because they answer a narrower question. The toy target in Table 5 is the unregularized behavior-policy gradient of terminal reward plus token cost. VIMPO, by contrast, is derived from a KL-regularized value identity. Its actor coefficient is meaningful inside that regularized objective and alongside the value-consistency loss used by the full algorithm. Comparing only the actor coefficient against the unregularized toy gradient is therefore a diagnostic of alignment with the toy target, not a test of full VIMPO.

This distinction matters in two directions. First, when $\pi_\theta = \pi_{\mathrm{ref}}$, the actor coefficient in Equation 8 is zero even though the ordinary reward gradient need not be zero. That fact should not be read as "VIMPO cannot learn at initialization," because the complete method can also include value-consistency terms and is not optimizing the same target as the toy. Second, when the reference is fixed at nonzero KL distance, the actor coefficient becomes token-varying and can be partially aligned with the ordinary gradient. That alignment is reference-sensitive: it depends on how the reference differs from the current policy, not only on the environment's reward structure.

A fair full-algorithm comparison would require a different audit. The target would be a KL-regularized objective such as

$$J_\beta(\theta) = \mathbb{E}_{\pi_\theta}[R(\tau)] - \beta\,\mathbb{E}_s[D_{\mathrm{KL}}(\pi_\theta(\cdot \mid s) \,\|\, \pi_{\mathrm{ref}}(\cdot \mid s))]\,, \tag{37}$$

with the same state weighting convention used by the implemented loss. The measured gradient would then be the gradient of the full objective, and the algorithmic update would include both the actor coefficient and the value-consistency term. This paper does not claim to have run that comparison. Instead, it uses the policy-implied rows to make a taxonomy point: critic-free does not imply trajectory-constant, but policy-implied token variation has its own reliability condition.

D Why Gradient Cosine and Token Credit Diverge

The exact-gradient audit gives a useful warning for interpreting credit assignment studies. A terminal-return coefficient can be a good Monte Carlo policy-gradient coefficient even if it is a poor local explanation of which token helped. The reason is that a policy-gradient update aggregates score-weighted coefficients:

$$\widehat{g} = \sum_t \widehat{A}_t \psi_t, \qquad \psi_t = \nabla_\theta \log \pi_\theta(a_t \mid o_t). \tag{38}$$

If two errors lie mostly in directions where ψ_t is small or cancel across actions, the gradient direction can remain good while token-level Pearson correlation is modest. Conversely, an estimator can correlate with local advantages but still produce a poor update if its errors are concentrated on high-score-norm actions.

For this reason, the paper reports several diagnostics rather than a single score. Token-level correlation asks whether the estimator explains local behavior-policy action credit. Calibrated MSE asks whether an affine rescaling can recover the target. Trace variance asks how noisy the sampled gradient is. Matched-KL improvement asks whether the mean update direction improves the exact finite-MDP return under a comparable local policy move. The score-weighted residual moment in Table 7 is closest to the variance term that a baseline directly controls.

This also clarifies why the broadcast ceiling is not a proof that trajectory-level methods cannot optimize. The ceiling is an information bound on token-local explanation under the chosen token weighting. A trajectory broadcast may still produce a useful update when the score-weighted projection of local advantages is well represented by trajectory context. The scientific question is therefore not whether group baselines are "valid" in the abstract. It is whether their information loss is acceptable for the horizon, heterogeneity, and action-score geometry of the task at hand.

Distillation losses. The dense distillation losses in the main text are written as KL objectives because the teacher signal is a distribution over next tokens. Other implementations may use sequence-level filtering, hidden-state matching, preference-filtered teacher traces, or mixed supervised/RL objectives. These variants can be useful without changing the taxonomy: they are dense imitation or self-imitation signals rather than direct policy-gradient reward optimization.

E Toy Environment Details

State. The state contains prompt identity, hidden progress, remaining horizon, and the features visible to the critic variant. The full critic observes enough to distinguish progress states. The coarse critic sees compressed state. The blind critic intentionally loses the feature needed for reliable future-return prediction.

Actions. The action set is deliberately small. `help` increases progress, `harm` decreases it, and `wait` consumes decision horizon without changing progress. This small action space makes exact dynamic programming feasible and makes the meaning of credit assignment visible. A language-model trace has richer tokens, but the policy-relative categories are recognizable: helpful substeps, harmful detours, delay actions, and formatting or null-like steps that future work should model separately.

Reward. The terminal verifier is sparse and thresholded. This makes the group-relative estimator plausible, because rollouts can be ranked by final success, but also creates the credit problem: the reward is delayed until after many actions.

Behavior-policy advantage. The behavior-policy advantage uses exact values under known dynamics. This is the scientific anchor of the paper. Without it, a method comparison could only say which estimator produced higher downstream score under one optimizer. With it, we can ask whether an estimator points in the same direction as the true policy-relative action credit.

F Observability and Privilege Controls

The main toy policy is not an LLM policy, but the observability distinction is still important. The behavior policy samples actions from threshold, progress score, and remaining horizon. The non-privileged critic uses the same information. Thus, in the non-privileged rows, a critic advantage is not winning because it sees hidden simulator state that the actor lacks; it is winning because it uses a state-conditioned temporal-difference construction rather than a token-invariant trajectory broadcast.

The coarse and blind critics are deliberately weaker controls. The coarse critic keeps the threshold and remaining horizon but compresses progress to its sign. The blind critic keeps threshold and remaining horizon but removes progress entirely. These modes mimic practical failures where a value model's input omits the causal feature needed for future return. In real language agents, this can happen through lossy compaction, missing tool state, hidden environment variables, or a value model that sees a sanitized transcript rather than the full executable state.

This is why the phase grid reports both observability and coverage. A full-information but undercovered critic can be unreliable because it has too few examples. A well-covered but blind critic can be unreliable because the state abstraction is wrong. A non-privileged and well-covered critic is the cleanest test of the value-estimation mechanism. The group-favorable rows are therefore not accidents to hide; they identify regimes where terminal trajectory information is more useful than a weak value estimate.

The same distinction should carry into future sequence-policy experiments. A fair LLM critic comparison should specify whether the critic sees exactly the actor's token history, a richer tool or environment state, process labels, hidden verifier state, or future-context teacher information. Centralized training with privileged state can be scientifically useful, but it should be labeled as such and compared separately from actor–critic training with the same observation.

G Case Design and Artifact Mapping

The 18 fixed cases are grouped by axis. Horizon cases increase trace length and delay-action exposure. Group-size cases vary the number of completions used for within-prompt normalization while keeping critic replay fixed at 840 training trajectories for every group-size row. Critic-budget cases vary how much data the value table sees before evaluation. Observability cases degrade the state available to the critic. Sparse-reward cases make terminal success harder. The counterexample combines blindness and low coverage.

The submission PDF reports selected rows and aggregate readings, while complete tables live in the repository artifacts. This split is intentional. The PDF should carry the mechanism argument, not raw run logs; the repository should carry enough machine-readable detail for independent inspection. The canonical deep matrix is stored as `results/deep_matrix_20seed.json` and `results/deep_matrix_20seed.csv`. The complete phase grid is stored as `results/credit_phase_diagram_seedset.json`. The policy-gradient audit, closed-loop audit, neural generalization audit, and robustness sweeps each have matching JSON and Markdown summaries under `results/`.

The result artifacts are generated, not hand edited. The build script reads the JSON files, regenerates the LaTeX tables used in the paper, compiles the PDF, and writes a manifest with SHA-256 hashes and page-count checks. This makes the paper auditable while avoiding pages of ledger-style tables in the main artifact.

H Benchmark Reporting Protocol

Trace schema. A real long-horizon benchmark should record more than a final reward. Each trajectory should include prompt identity, sampled response identity, token or step indices, tool calls, tool observations, compaction boundaries, verifier calls, process labels when available, terminal reward, and any safety or refusal events. This schema is the minimum needed to compare a group-relative estimator with a value-model estimator after the fact. Without it, the experiment can still report final success, but it cannot diagnose where credit was assigned or lost.

Budget matching. The clean comparison is not "PPO run" versus "GRPO run" in isolation. The comparison should state generated-token budget, number of prompts, completions per prompt, verifier calls, critic update steps, policy update steps, and peak activation or optimizer memory. A critic-based method should be charged for critic fitting and storage. A group-relative method should be charged for the extra completions needed to obtain reward contrast. A distillation method should be charged for teacher inference or self-teacher selection.

Primary measurements. For any benchmark with exact or process annotations, report estimator correlation with behavior-policy credit, calibrated MSE, sign accuracy, zero-variance group fraction, and return improvement under equal rollout budgets. For benchmarks without exact credit, report held-out process-label agreement, per-step value calibration where labels exist, and sensitivity to trajectory length. Final score alone is not enough because a method can improve a short task while failing the mechanism required for long traces.

Decision rule. A claim that one family is better should require at least three checks: the mean effect favors the method, the confidence interval excludes practical near-ties, and an adversarial or counterexample regime is reported. This paper's toy result follows that rule: it highlights the critic-favorable mean pattern, the near tie under tiny critic budget, and the group-favorable blind undercovered case. A larger benchmark should preserve the same discipline.

Audit package. The artifact package should include raw per-seed rows, aggregate tables, plotting scripts, exact commands in repository documentation, environment notes, and a short statement of non-claims. The non-claims matter. For this line of work, the honest claim is that critic-style temporal information can dominate in observable long traces, while group-relative normalization remains strong when reward contrast is reliable and value modeling is weak. That is a research program, not a slogan.

I Minimal Sequence-Policy Benchmark Specification

The next external-validity step should be small enough to run repeatedly and large enough to remove the main weakness of the current paper: table-lookup state. A useful benchmark can encode the same finite-MDP family as token sequences and train a tiny autoregressive policy. Each prompt specifies a threshold, horizon, and optional hidden distractor. The policy emits action tokens from a small vocabulary such as `help`, `harm`, `wait`, and `stop`; the environment then appends an observation token containing remaining horizon and a compressed progress observation. The verifier returns a terminal success bit and optional process labels for selected prefixes.

Policy and value models. The minimal model should be a one- or two-layer transformer with shared token embeddings and separate policy and value heads. The value head should be tested in three access modes: actor-history only, actor-history plus explicit environment state, and deliberately blinded history. The first mode is the fair non-privileged comparison. The second mode is centralized training with richer state and must be reported separately. The third mode is the critic-failure control.

Training arms. At equal generated-token budgets, compare at least five arms:

1. response-level sibling leave-one-out or simplified GRPO;
2. actor–critic TD or GAE with the non-privileged value head;
3. sampled Monte Carlo values from prefix branching;
4. a VIMPO-style policy-implied actor coefficient evaluated inside a matched KL-regularized objective if implemented as a full update;
5. an anchor or prefix baseline when repeated states or intermediate verifier calls exist.

The experiment should include corrected GRPO-family variants only after the simple sibling baseline is stable. Otherwise optimizer differences will obscure the credit mechanism.

Budget matching. Every arm should report generated tokens, verifier calls, reference-policy scoring calls, value-model forward/backward passes, sampled branches, wall-clock time, and peak memory. A critic method should not be called better if it wins only by silently using more environment interaction. A group method should not be called cheaper if it spends many discarded completions to obtain reward contrast.

Measurements. Because the environment remains finite, the benchmark can still compute exact or high-precision behavior-policy advantages for sampled prefixes. Report token-credit Pearson correlation, calibrated MSE, gradient cosine with the exact finite-MDP gradient when available, matched-KL one-step improvement, learning curves over at least five seeds, final return, success rate, length, delay-token fraction, and estimator-selection regret. The key result would be whether the phase boundary observed in the tabular audit survives neural function approximation and closed-loop policy learning.

Required ablations. The benchmark should include ablations that can falsify the paper's mechanism claim. First, remove delay actions while holding terminal reward sparsity fixed; trajectory broadcasts should become more competitive because within-trajectory credit heterogeneity is lower. Second, keep delay actions but hide progress from the value head; the critic should lose reliability even though the task is unchanged. Third, increase sibling group size while holding critic replay fixed; this isolates group contrast from value-model coverage. Fourth, add a cheap prefix verifier at one or two intermediate budgets; prefix baselines should reduce score-weighted residual variance even if they do not become the best token-credit estimator. Fifth, run a high-branch sampled-value arm on a small subset of prefixes; this provides a critic-free estimate of the value signal that can calibrate whether the learned critic is underfitting or whether the state information itself is insufficient.

Acceptance criteria. A neural follow-up should be considered supportive only if it passes both an optimizer test and an estimator test. The optimizer test asks whether the method improves return under matched generated-token and verifier budgets. The estimator test asks whether the credit signal aligns with exact or audited behavior-policy advantages on held-out prefixes. A method that wins the optimizer test but fails the estimator test may still be useful, but it is not evidence for the credit-assignment mechanism claimed here. Conversely, a method that wins the estimator test but loses the optimizer test identifies an engineering or cost bottleneck rather than disproving the mechanism.

J Estimator-Selection Procedure

The paper's practical recommendation can be written as an explicit audit procedure rather than a slogan. Given a candidate task distribution, collect a small diagnostic batch before launching a large post-training run. For each trajectory, store token indices, prompt identity, terminal reward, length, available process labels, value predictions, group statistics, and any prefix verifier calls. Then estimate three quantities:

$$\widehat{H}_{\text{credit}} = \frac{\widehat{\mathbb{E}}\left[\widehat{\mathrm{Var}}\left(\widehat{A}^{\text{audit}}_{i,t} \mid C_i\right)\right]}{\widehat{\mathrm{Var}}\left(\widehat{A}^{\text{audit}}_{i,t}\right)}, \tag{39}$$

$$\widehat{R}_V = \mathrm{Corr}(\widehat{A}^V_t, \widehat{A}^{\text{audit}}_t), \tag{40}$$

$$\widehat{C}_m = (\text{tokens}_m, \text{verifier}_m, \text{memory}_m, \text{time}_m). \tag{41}$$

Here $\widehat{A}^{\text{audit}}_t$ may come from exact simulation, sampled continuations, trusted process labels, or a small counterfactual audit. It need not be available for every production token; it only has to be good enough to rank estimator reliability on a held-out diagnostic slice. If $\widehat{A}^{\text{audit}}$ is estimated by sampled continuations, raw within-trajectory variance is upper-biased by Monte Carlo noise; repeated branches should be used to estimate and subtract continuation noise, or the reported $\widehat{H}_{\text{credit}}$ should be labeled an upper-biased diagnostic.

Decision rule. A simple conservative policy is:

$$m^\star = \arg\min_m \left[\widehat{\mathrm{MSE}}_{\text{audit},m} + \lambda^\top \widehat{C}_m\right], \tag{42}$$

where

$$\widehat{\mathrm{MSE}}_{\text{audit},m} = \widehat{\mathbb{E}}\left[\left(\widehat{A}^m_{i,t} - \widehat{A}^{\text{audit}}_{i,t}\right)^2\right]. \tag{43}$$

This deployable rule folds information loss and reliability error into one held-out audit loss and then charges resources separately. In the toy, trajectory broadcasts have high audit MSE when H_{credit} is large; critics have high audit MSE when coverage or observability is poor; structural methods have high audit MSE when anchor support is sparse; sampled values have high cost when branch budgets are large. The best method is the one whose lower audit error justifies its resource cost. Table 9 reports a small held-out version of this procedure on the phase grid.

Reporting rule. Do not publish only the best final score. Publish the diagnostic batch, estimator-selection rule, all negative controls, and at least one counterexample regime. If a critic wins, show that it is not using privileged information unless that is the intended setting. If a group method wins, show that reward dispersion and group comparability are sufficient. If a policy-implied method wins, specify the reference policy, KL target, and whether the reported update is a full algorithmic objective or only an actor-coefficient diagnostic. This is the difference between a leaderboard claim and a reusable scientific result.

References

[1] Rishabh Agarwal et al. On-policy distillation of language models: Learning from self-generated mistakes, 2023. URL https://arxiv.org/abs/2306.13649.

[2] Arash Ahmadian, Chris Cremer, Matthias Galle, Marzieh Fadaee, Julia Kreutzer, Olivier Pietquin, Ahmet Ustun, and Sara Hooker. Back to basics: Revisiting reinforce style optimization for learning from human feedback in llms, 2024. URL https://arxiv.org/abs/2402.14740.

[3] Jose A. Arjona-Medina, Michael Gillhofer, Michael Widrich, Thomas Unterthiner, Johannes Brandstetter, and Sepp Hochreiter. Rudder: Return decomposition for delayed rewards, 2018. URL https://arxiv.org/abs/1806.07857.

[4] DeepSeek-AI. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning, 2025. URL https://arxiv.org/abs/2501.12948.

[5] Lang Feng, Zhenghai Xue, Tingcong Liu, and Bo An. Group-in-group policy optimization for llm agent training, 2025. URL https://arxiv.org/abs/2505.10978.

[6] Wei Fu, Jiaxuan Gao, Xujie Shen, Chen Zhu, Zhiyu Mei, Chuyi He, Shusheng Xu, Guo Wei, Jun Mei, Jiashu Wang, Tongkai Yang, Binhang Yuan, and Yi Wu. Areal: A large-scale asynchronous reinforcement learning system for language reasoning, 2025. URL https://arxiv.org/abs/2505.24298.

[7] Yiran Guo, Lijie Xu, Jie Liu, Dan Ye, and Shuang Qiu. Segment policy optimization: Effective segment-level credit assignment in rl for large language models, 2025. URL https://arxiv.org/abs/2505.23564.

[8] Jian Hu, Jason Klein Liu, Haotian Xu, and Wei Shen. Reinforce++: Stabilizing critic-free policy optimization with global advantage normalization, 2025. URL https://arxiv.org/abs/2501.03262.

[9] Zhewei Kang, Aosong Feng, Sergey Levine, Dawn Song, and Xuandong Zhao. Vimpo: Value-implicit policy optimization for llms, 2026. URL https://arxiv.org/abs/2606.20008.

[10] Amirhossein Kazemnejad, Milad Aghajohari, Eva Portelance, Alessandro Sordoni, Siva Reddy, Aaron Courville, and Nicolas Le Roux. Vineppo: Refining credit assignment in rl training of llms, 2024. URL https://arxiv.org/abs/2410.01679.

[11] Azal Ahmad Khan, Ammar Ahmed, Zeshan Fayyaz, Sheng Di, Mingyi Hong, and Ali Anwar. Faster synchronous on-policy rl via straggler-aware group sizing, 2026. URL https://arxiv.org/abs/2606.02218.

[12] Jiazheng Li, Yawei Wang, David Yan, Yijun Tian, Zhichao Xu, Huan Song, Panpan Xu, and Lin Lee Cheong. Salt: Step-level advantage assignment for long-horizon agents via trajectory graph, 2025. URL https://arxiv.org/abs/2510.20022.

[13] Yaxuan Li, Yuxin Zuo, Bingxiang He, Jinqian Zhang, Chaojun Xiao, Cheng Qian, Tianyu Yu, Huan-ang Gao, Wenkai Yang, Zhiyuan Liu, and Ning Ding. Rethinking on-policy distillation of large language models: Phenomenology, mechanism, and recipe, 2026. URL https://arxiv.org/abs/2604.13016.

[14] Yu Li, Rui Miao, Tian Lan, and Zhengling Qi. Oppo: Bayesian value recursion for token-level credit assignment in llm reasoning, 2026. URL https://arxiv.org/abs/2605.21851.

[15] Yuanfan Li, Qi Zhou, Wenjing Duan, and Lu Chen. When denser credit is not enough: Evidence-calibrated policy optimization for long-horizon llm agent training, 2026. URL https://arxiv.org/abs/2606.05885.

[16] Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step, 2023. URL https://arxiv.org/abs/2305.20050.

[17] Zichen Liu et al. Understanding r1-zero-like training: A critical perspective, 2025. URL https://arxiv.org/abs/2503.20783.

[18] Feng Luo, Yu-Neng Chuang, Guanchu Wang, Zicheng Xu, Xiaotian Han, Tianyi Zhang, and Vladimir Braverman. Demystifying opd: Length inflation and stabilization strategies for large language models, 2026. URL https://arxiv.org/abs/2604.08527.

[19] Yifu Luo, Zeyu Chen, Haoyu Wang, Xinhao Hu, Yuxuan Zhang, Zhizhou Sha, and Shiwei Liu. Learning from the self-future: On-policy self-distillation for dllms, 2026. URL https://arxiv.org/abs/2606.18195.

[20] Youssef Mroueh. Reinforcement learning with verifiable rewards: Grpo's effective loss, dynamics, and success amplification, 2025. URL https://arxiv.org/abs/2503.06639.

[21] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback, 2022. URL https://arxiv.org/abs/2203.02155.

[22] Penghui Qi, Zichen Liu, Tianyu Pang, Chao Du, Wee Sun Lee, and Min Lin. Optimizing anytime reasoning via budget relative policy optimization, 2025. URL https://arxiv.org/abs/2505.13438.

[23] Mohammad Mahdi Salmani-Zarchi et al. Mdp-grpo: Stabilized group relative policy optimization for multi-constraint instruction following, 2026. URL https://arxiv.org/abs/2606.06058.

[24] John Schulman, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel. High-dimensional continuous control using generalized advantage estimation, 2015. URL https://arxiv.org/abs/1506.02438.

[25] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms, 2017. URL https://arxiv.org/abs/1707.06347.

[26] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models, 2024. URL https://arxiv.org/abs/2402.03300.

[27] Nisan Stiennon, Long Ouyang, Jeffrey Wu, Daniel Ziegler, Ryan Lowe, Chelsea Voss, Alec Radford, Dario Amodei, and Paul Christiano. Learning to summarize with human feedback, 2020. URL https://arxiv.org/abs/2009.01325.

[28] Peiyi Wang, Lei Li, Zhihong Shao, R. X. Xu, Damai Dai, Yifei Li, Deli Chen, Y. Wu, and Zhifang Sui. Math-shepherd: Verify and reinforce llms step-by-step without human annotations, 2023. URL https://arxiv.org/abs/2312.08935.

[29] Zhongwen Xu and Zihan Ding. Single-stream policy optimization, 2025. URL https://arxiv.org/abs/2509.13232.

[30] Fengkai Yang, Zherui Chen, Xiaohan Wang, Xiaodong Lu, et al. Your group-relative advantage is biased, 2026. URL https://arxiv.org/abs/2601.08521.

[31] Qiying Yu et al. Dapo: An open-source llm reinforcement learning system at scale, 2025. URL https://arxiv.org/abs/2503.14476.

[32] Lifan Yuan, Wendi Li, Huayu Chen, Ganqu Cui, Ning Ding, Kaiyan Zhang, Bowen Zhou, Zhiyuan Liu, and Hao Peng. Free process rewards without process labels, 2024. URL https://arxiv.org/abs/2412.01981.

[33] Yu Yue, Yufeng Yuan, Qiying Yu, et al. Vapo: Efficient and reliable reinforcement learning for advanced reasoning tasks, 2025. URL https://arxiv.org/abs/2504.05118.

[34] Z.ai. Glm-5.2: Built for long-horizon tasks, 2026. URL https://z.ai/blog/glm-5.2. Vendor report, accessed June 18, 2026.

[35] Chenchen Zhang. From reasoning to agentic: Credit assignment in reinforcement learning for large language models, 2026. URL https://arxiv.org/abs/2604.09459.

[36] Siyan Zhao, Zhihui Xie, Mengchen Liu, Jing Huang, Guan Pang, Feiyu Chen, and Aditya Grover. Self-distilled reasoner: On-policy self-distillation for large language models, 2026. URL https://arxiv.org/abs/2601.18734.

[37] Hongyi Zhou et al. Demystifying group relative policy optimization: Its policy gradient is a u-statistic, 2026. URL https://arxiv.org/abs/2603.01162.

[38] Yifei Zhou, Andrea Zanette, Jiayi Pan, Sergey Levine, and Aviral Kumar. Archer: Training language model agents via hierarchical multi-turn rl, 2024. URL https://arxiv.org/abs/2402.19446.